Lab 1 → Access & tour AWS console

AWS ≠ website/server
= massive global operating system for infrastructure

AWS Management Console = remote control panel

Regions = physically seperate cloud data centers across the world
Services = building blocks (compute, storage, DB, networking)
IAM user login = controlled access to a tiny slice of that universe

AWS management console → 1 way to talk to AWS (others → CLI, SDK, API's)
- → web based UI
- → Backed by AWS control plane API's
- → used to configure, not directly run, infra

↓
Access AWS control plane via management console to manage regional & global AWS resources

IAM answers 3 core questions
- → who are you? (authentication)
- → what can you do? (authorization)
- → on which resources? (scope)

IAM users ✓ root ✗
↓
god mode + dangerous + has unrestricted access, even without policies

Sign in ⇒ what actually happens internally?
1. Password verified by IAM
2. Session token issued
3. Browser stores temp. session
4. Console UI loads based on permissions

This is control plane authentication, not data plane

Region → geographic location
↓
contains many AZ
↓
completely isolated from other regions

I am choosing where my resources live physically (OK)

I am scoping my control plane operations to US-east-1 endpoint

US-east-1 ⇒ oldest region
⇒ cheapest for many services
⇒ most services launch here 1st

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If wrong region → you want see your resources
↳ API return "resource not found"
↳ Debugging nightmares

EC2 → regional
S3 → global namespace, regional data
IAM → global

when you change regions → console reloads, same UI, diff backend endpoints

disabled regions → AWS disables some regions → compliance / legal restrictions / billing controls / not enabled for your acc.
⇓
Companies often:
↳ Restrict regions via Service control policies
↳ Enforce data residency laws

AWS console mobile App → AWS supports multi-device mgmt
↓
mobile control plane client

what it can do
⇓
- Monitor EC2
- Start/stop instances
- view cloudwatch alarms

what it can't do
⇓
- complex architecture changes
- large deployments

★ Control Plane vs Data Plane

Control Plane → deciding what should happen
⇓
It decides:
- which plane flies
- where it flies
- who is allowed onboard

↳ if scheduling comp. goes down → The plane that are already flying keep flying

Data Plane → actually making it happen
⇓
- The actual airplane
- passengers
- luggage

SurbhiExplains

Control plane

- Set of AWS managed API's
- Running on AWS's internal infrastructure
- Responsible for resource mgmt.

→ we dont see the control plane directly → we only interact through it.

↓

we interact with control plane by: mgmt console + CLI + SDK + CloudFormation + Terraform

↓

all roads lead to same control plane API's

eg. Create an EC2 instance
Attach this IAM role
open port 443
Scale ASG to 5

none of this moves real user traffic

↓

Its instructions & metadata

BTS → Click launch EC2 instance

1. Browser sends request to AWS API endpoint
2. AWS authenticates via IAM
3. AWS validates → Quotas, permissions, Region
4. AWS records desired state

↓

"This account wants 1 EC2 instance of type t3.micro"

5. control plane schedules data plane resources
6. Returns success response to you

At this point → instance may not even be running yet
BUT the decision is locked in!

Control plane defines Intent not execution

Data plane

↳ where your code runs + packets flow + users connect

⇓

real workload execution

SubhiExplains

eg. of Data plane activity

⇓

- HTTP request hitting an EC2 instance
- λ fxn executing code
- S3 serving an object
- RDS processing a SQL query

} No configuration here → just work

AWS Console is down in a region. will my application stop running?

↓

NO! As long as → data plane = healthy
↳ resources already exist

Because → console = control plane
↳ App traffic = data plane

Control plane IAM ⇒ Can this user create EC2?
↳ Can this role attach policy?

Data plane IAM ⇒ Can this lambda read from S3?
↳ Can this EC2 access dynamoDB?

} Same IAM service
Diff. plane
Diff consequences

Real world Production Example

⇓

Netflix

Control plane

- deploys new EC2 fleets
- updates autoscaling policies
- rotates IAM roles

⇓

If this goes down
- Engineers panic
- users keep watching

Data plane

- millions of video streams
- API requests
- user authentication flow

⇓

If this goes down:
- Twitter explodes
- Engineers cry

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Cost Implications

- Control plane actions are mostly free
  - Creating resources
  - modifying configs
- Data plane usage is what you pay for
  - compute hours
  - data transfer
  - requests

Security Implications

- Lock down control plane
  - least privilege IAM
  - MFA
  - SCP's
- Protect data plane
  - Network security
  - encryption
  - runtime permissions

Many breaches happen because: control plane permissions were too open

You → console/CLI/SDK
↓
CONTROL PLANE (API's, IAM, config)
↓
DATA PLANE (EC2, lambda, S3, Traffic)
↓
End users

end users talk to the data plane
↓
Importantly ⇒ Most data plane traffic never touches the AWS console/IAM API's

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